GlucoTwin live platform

Implementation strategy, current evidence and the path to authorized clinical use

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Product decision

Retain GlucoTwin and organize delivery into four workstreams. Its distinctive experience connects an arriving reading to a permanent forecast, a clinician decision and the eventual outcome. The model's changing state remains inspectable.

What this release enables

A redesigned clinician review desk, patient evidence workspace, patient companion and clinic administration. Synthetic readings enter durable jobs automatically; a worker saves an arrival-aware research receipt and updates the queue. Recorded public-data cases provide reproducible future-outcome demonstrations.

Challenge need	Concrete evidence
Historical plus sensor input	Dated context and native CGM readings; separate synthetic patient identities.
Dynamic patient model	Population model with bounded correction from matured, actually issued forecasts.
Upcoming event	Measurement-anchored glucose-excursion research forecasts; target clock and remaining lead shown.
Doctor interaction	Saved evidence, review disposition, follow-up, approved sharing and consultation PDF.

Source status

Clinic connection pending. No CGM device or authorized clinic stream has been supplied. A server-driven synthetic stream is implemented and labeled. Genuine wearable data, prospective clinical validation and clinical release remain separate gates.

The clinician experience

One review episode connects observed data, issued evidence and care-team responsibility

Workspace	Behavior and acceptance
Review desk	Search and filter responsibility/status/data issues; show review age, freshness and current source status.
Patient evidence	Inspect trajectory, forecast comparison, uncertainty and input cutoff. A saved receipt stays selected when new observations arrive.
Review action	Record disposition and follow-up. Expected revision prevents concurrent reviewers from silently overwriting one another.
Timeline	Align native readings, dated context, issued forecasts, observed outcomes and actions. Keep human and replay clocks distinct.
Evidence explanation	Show measured trend, model correction, maturity and missing inputs as model associations.
Consultation report	Export the selected evidence period and unresolved work with provenance and gap restrictions.

Visual system

Warm white, navy and teal; amber for research review, violet for comparison, grey for unavailable evidence. Self-hosted Inter, visible keyboard focus, accessible dialogs, text summaries and reading tables accompany the chart.

Prediction and workflow findings

Useful warning, non-actionable warning, data problem and further assessment are clinician workflow dispositions. Event occurrence is computed separately from measured outcomes. Acknowledgement cannot improve forecast accuracy.

Patient and administrator

A practical companion and an operational control desk

Product	Implemented scope
Patient companion	Own-record readings, self-reported meal/activity/symptom context, published follow-up, approved notes, consent history and revocation request.
Patient export	Patient-visible context and published instructions. Research predictions and private clinician assessment stay out of this report.
Installable web app	PWA manifest and static-only offline shell. Health API responses are not cached. An offline action waits for server acknowledgement.
Staff lifecycle	Invitations, single-use activation, deactivation/session revocation, roles and optional encrypted TOTP enrollment.
Source administration	Synthetic device-to-patient mapping, scoped keys, pause/revoke, ingestion/job health and explicit pending clinic source.
Intake	CSV preview with row-level errors/duplicate checks, accepted-count reporting and limited local FHIR adaptation.

Release boundary

Patient-facing predictions are disabled. Caregiver access, autonomous medication advice, treatment simulation and broad hospital connectivity are deferred. Administrative controls are research implementation evidence; they do not establish production privacy or healthcare approvals.

Arrival-aware processing

Separate sensor measurement from receipt, issuance and forecast target

Validated intake inserts the reading and durable job in one transaction. A leased worker uses only currently available information, saves a permanent receipt, projects the review episode and appends metadata to a transactional event outbox. Retry and input identity checks prevent duplicate issuance.

Time / compatibility	Rule
Measurement	Original native sensor timestamp/value preserved.
Receipt	Server establishes first trusted availability. Backfilled information cannot enter an earlier receipt.
Forecast	Server issue time, input cutoff, model/policy identity and actual target timestamps saved.
Delay example	10:00 measurement arriving at 10:01 retains target 10:30: 29 minutes remaining.
Event probability	Suppress when any original scheduled classifier target is already past; never rename it a new next-30-minute probability.
Cadence	Arrival adapter requires native 15-minute observations. Other cadences produce an explicit unsupported state. No invented observations.
Outcome	Endpoint result and complete-window event determination are separate append-only records.

Deployment structure

React/TypeScript, FastAPI and a separate forecasting worker; SQLite demonstration and explicit PostgreSQL migrations. No message broker is required. Authenticated server-sent events invalidate server-state caches without changing a selected forecast.

Model evidence and uncertainty

Internal retrospective performance is distinct from arrival engineering evidence

Evidence	Measured finding / interpretation
Five-fold historical model	30-minute MAE 9.606 mg/dL hybrid versus 9.812 population. Paired difference -0.206; patient-bootstrap 95% CI [-0.336,-0.110]. Small internal improvement.
Historical event evidence	Precision 62.96% [59.51,66.38]; overall recall 76.06% [73.82,78.67]. False alerts 0.919 [0.787,1.059] per 24 eligible forecast hours.
Arrival stress	20 patients; 60 sparse synthetic issuances per delay. 0/1/5-minute delays retain endpoint lead 30/29/25 minutes. 20-minute delay suppresses event probabilities.
Arrival adaptation	40 scored endpoints: population MAE 12.550 versus experimental hybrid 12.676. No arrival personalization gain established.
Scale exercise	50 synthetic sources; 400 accepted readings/jobs; 50 receipts and reviews; no failed jobs. Short local batch with actual model; not sustained device throughput.

Required next evidence

Obtain a permitted device source. Freeze an issuance-anchored evaluation with delay, cadence and missingness inputs. Evaluate exact deployment artifact, staffed queue burden, review delays, probability calibration, interval coverage, abstention and subgroup sample sizes.

Boundaries

The Shanghai cohort is Chinese hospital data. Its timezone is not independently verified for deployment. Synthetic delay and load tests do not establish clinical effectiveness, Indian generalization or real-device calibration.

Four coordinated workstreams

Delivery ownership and the hackathon calendar

Owner	Deliverable
A / Data and evidence	Arrival/time audit, calibration, cadence compatibility and policy replay.
B / Platform	Durable jobs, immutable receipts, migrations, scoped access, recovery and operations.
C / Product	Clinician workspace, companion/admin, responsive layout and accessibility checks.
D / Clinical and submission	Clinic source packet, doctor review, evidence ledger, architecture, presentation and video.

Dates, IST	Milestone
10-11 October	Confirm interfaces and clocks; prepare clinical-source assessment.
12-14 October	Reading-to-forecast-to-review flow and clinician redesign.
15-16 October	Concurrency/arrival tests, operational views, companion and doctor usability review when available.
17 October	Feature/model freeze; three fresh judge journeys.
18-19 October	Finalize public source, reports, editable presentation and at least 20-minute explanation.
20 October	Verify accessibility of assets; stop changes 5 PM IST before 7 PM deadline.

Clinic dependency

The participant approaches a clinician or college-linked clinic. No outreach has been sent. If access is delayed, independent platform work continues with explicit synthetic/recorded labels.

Clinical source and release gates

Progress from research platform to authorized collection, shadow study and supervised service

Stage	Exit evidence / current status
Research release	Automated pipeline, saved evidence, UI, access and local recovery checks. Public/synthetic records only.
Authorized collection	Clinic/device authorization, patient mapping, units, native cadence, timezone, delivery latency and permitted storage verified. Pending.
Shadow study	Approved intended-use protocol, frozen artifact, prespecified study size and workload/performance criteria. Pending.
Supervised care	Evidence accepted, applicable approvals, trained staffing, escalation and stop rules. Pending.

Operational work before private records

MFA/access review for all operational staff; verified TLS and encrypted storage; environment-managed secrets; consent/retention/withdrawal policy; redacted logs; always-on hosting; off-host encrypted backup/recovery and model rollback. Local encrypted backup tests do not establish an off-host recovery arrangement.

Three-minute flagship demonstration

Review desk -> incoming synthetic stream -> immutable forecast/input boundary -> population/adaptation comparison -> review and follow-up -> reproducible recorded outcome -> gap abstention -> consultation PDF. Every segment identifies source and clock. The long explanation adds architecture, evidence, limitations and reproducibility.

Submission

Public repository with README/team/stack/license; architecture and presentation PDF/PPT; explanation/demo video at least 20 minutes. Registration 18 October at 7 PM IST; prototype submission 20 October at 7 PM IST. Platform registration/submission remain participant-owned.

References and claim boundaries

Primary sources and local evidence records

01. Official challenge requirements and dates

02. Shanghai matched clinical and CGM data

03. Dexcom authorization process

04. ABDM FHIR R4 implementation guide

05. WCAG 2.2 accessibility reference

06. CDSCO Medical Device Software guidance

07. IMDRF clinical evaluation framework

08. MeitY DPDP publications

09. Public implementation and submission assets

Local records

docs/VALIDATION.md; docs/LIVE_MODEL.md; docs/OPERATIONS.md; data/processed/clinic_benchmark.json; data/processed/arrival_stress.json. Performance values refer to their documented cohorts and clocks.

Assumptions and pending evidence

Existing computer and free tools; four-workstream output; one clinic/device family first. No actual clinic access, doctor feedback, prospective study or clinical approval is assumed. No inferred success probability or official numerical judging weights.

Interoperability and accessibility

ABDM resources guide the limited local FHIR adapter; ecosystem connectivity/profile conformance are not established. WCAG 2.2 AA remains a target until complete automated and manual review; responsive checks alone do not certify conformance.